

**INDIAN INSTITUTE OF  
INFORMATION TECHNOLOGY  
GUWAHATI**

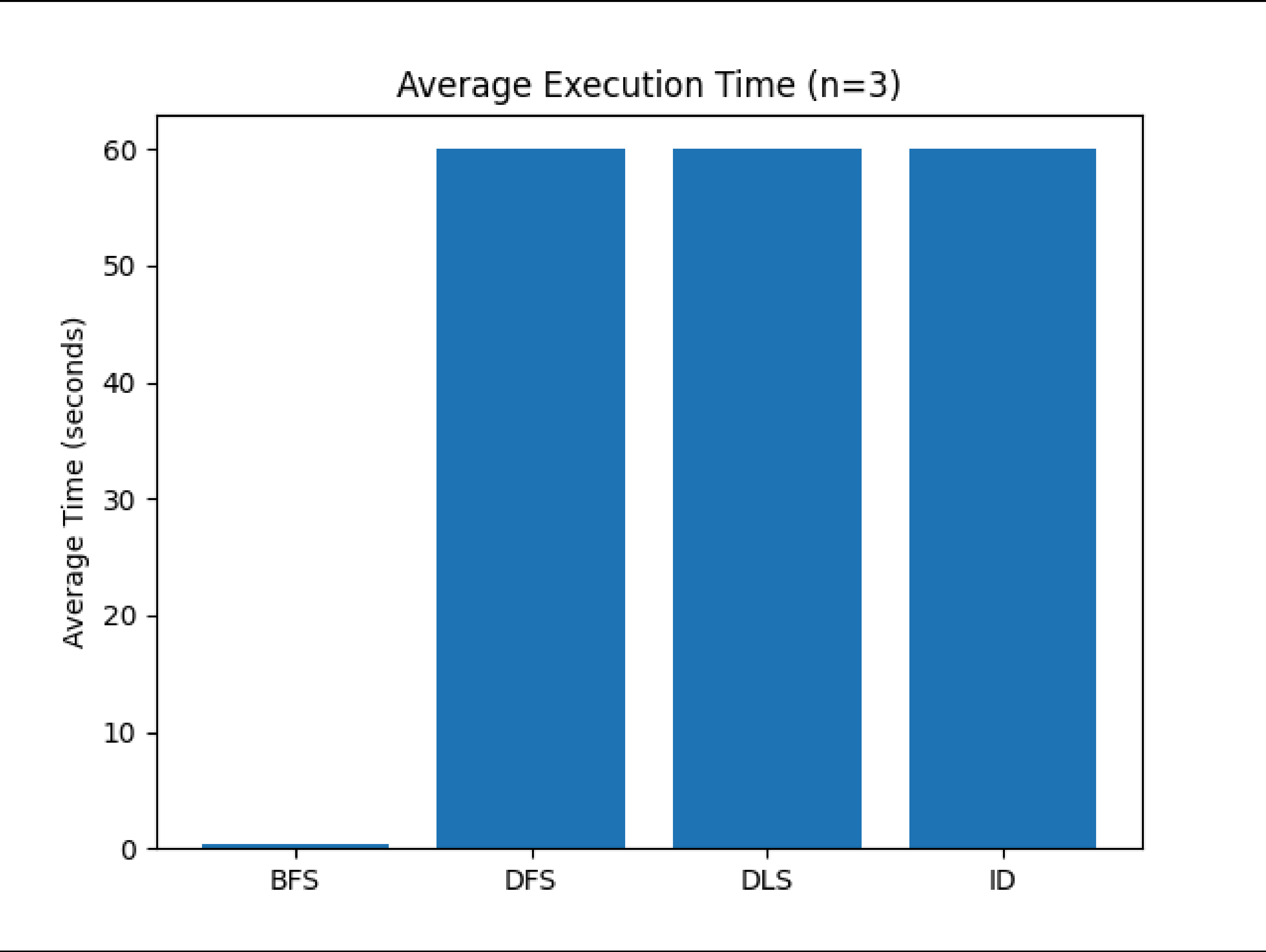


**AI LAB  
Assignment-4**

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**Course:     Artificial Intelligence Lab**  
**Code:        CS236**

# Average Execution Time

For n = 3

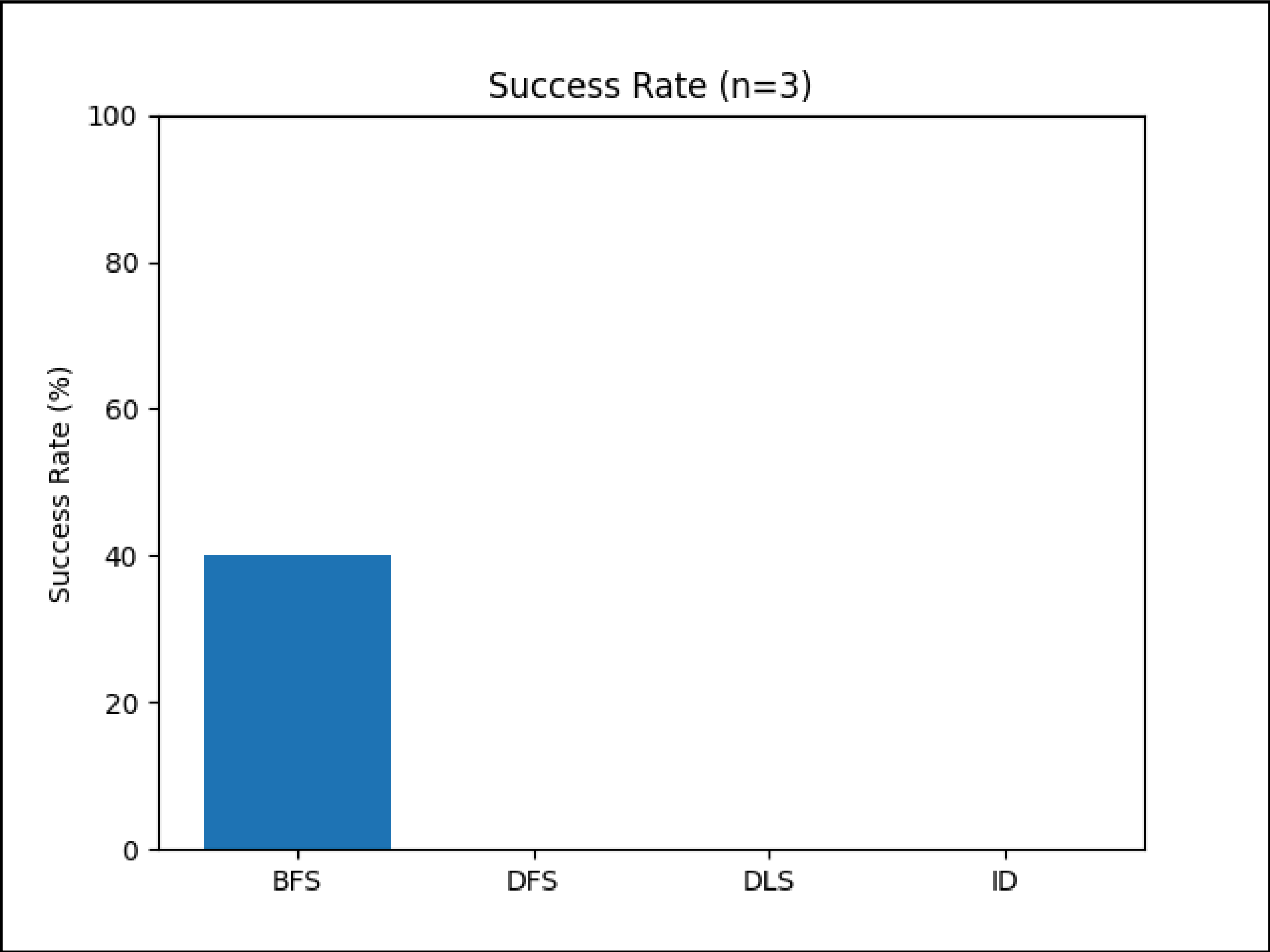


For n = 4

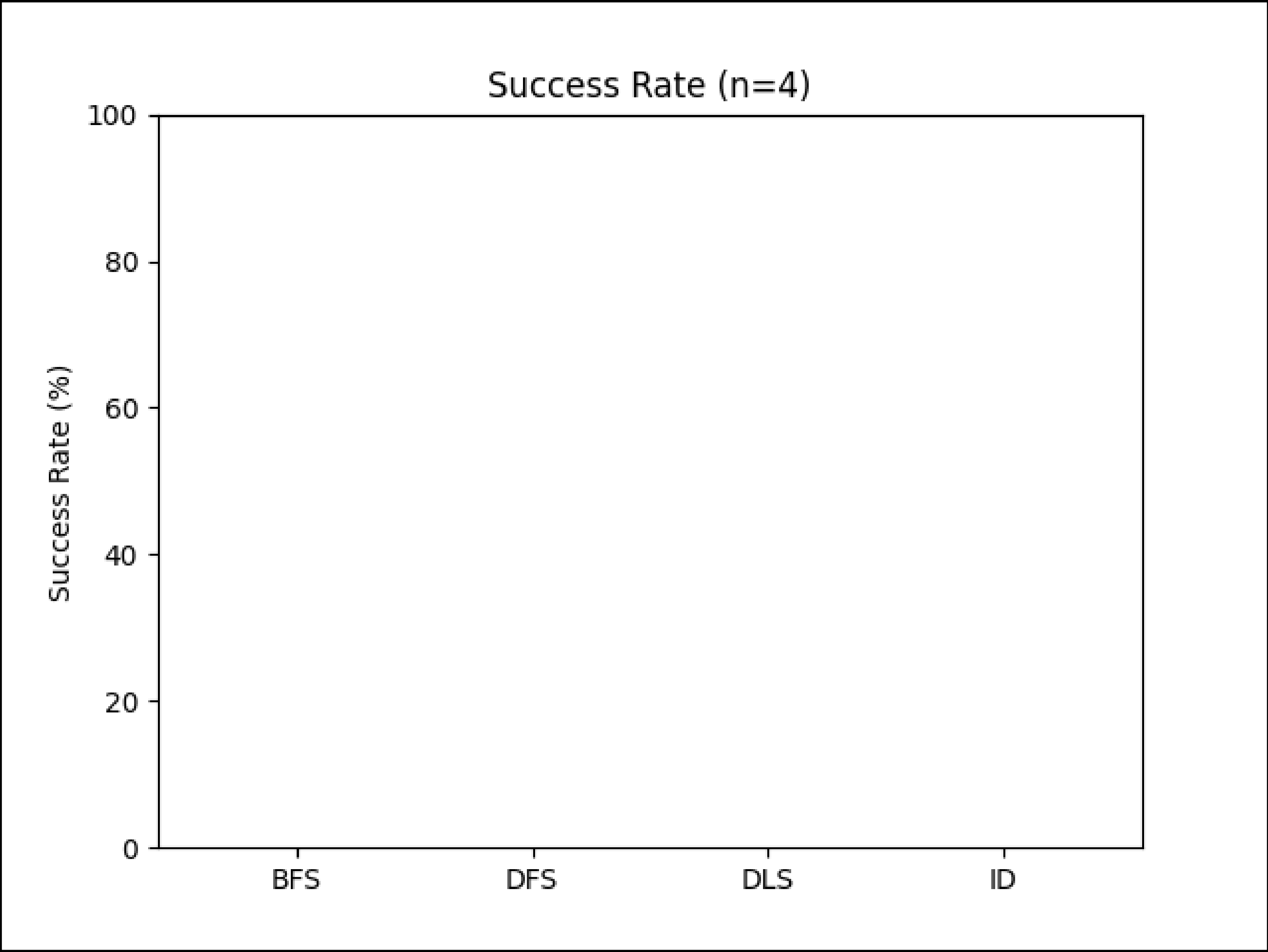


# Success Rate in Percent

For n = 3

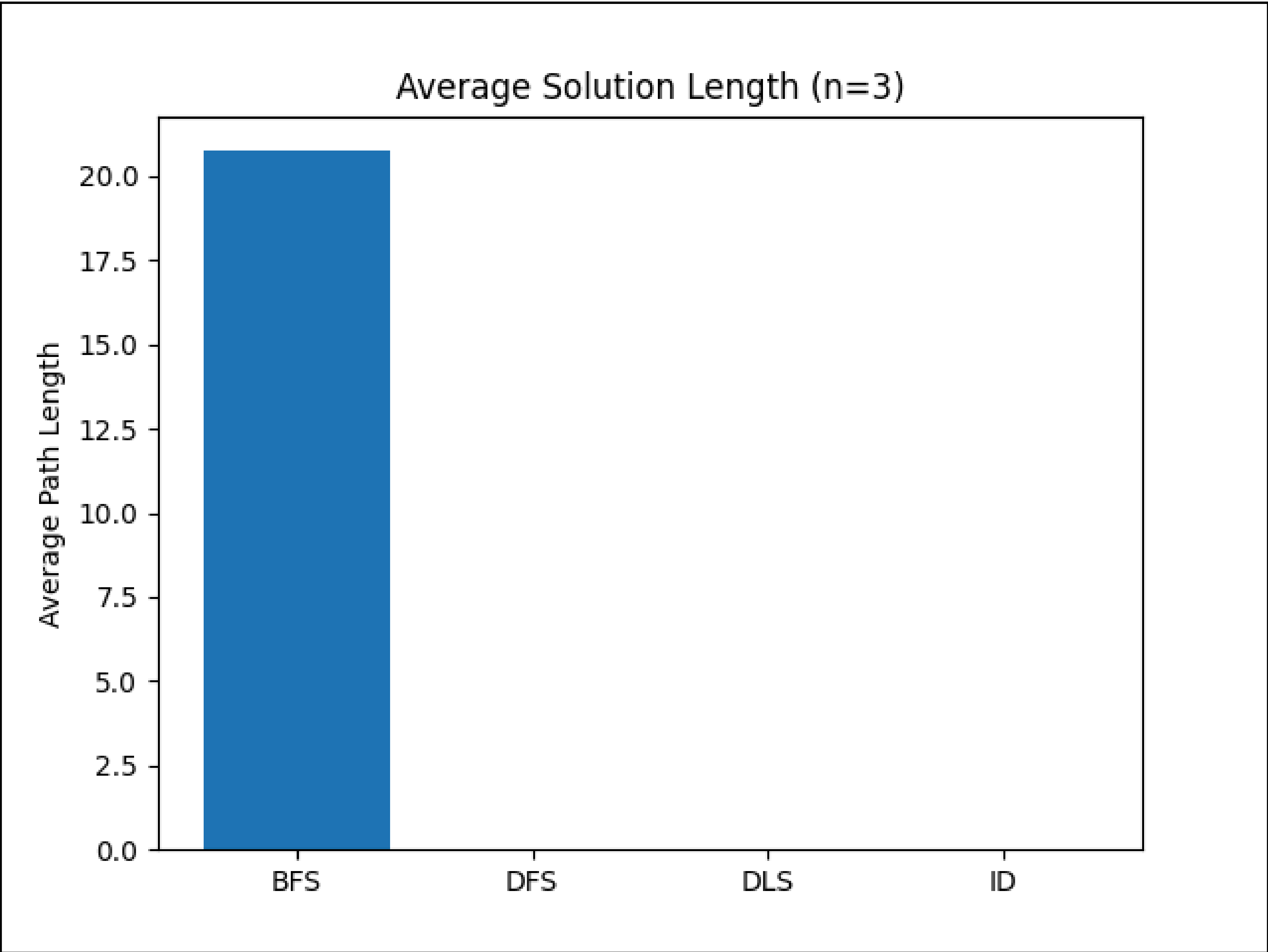


For n = 4

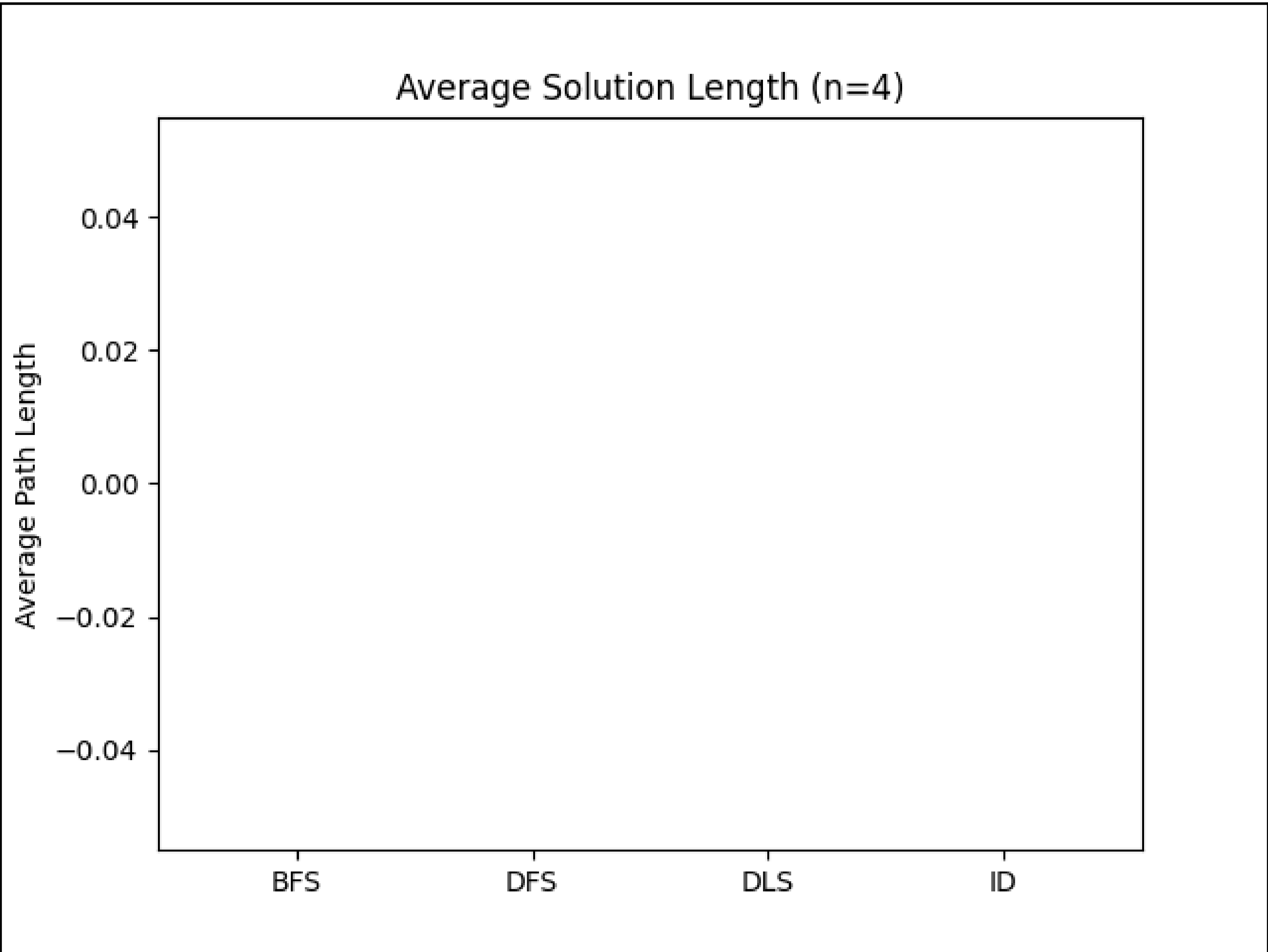


# Average Path Length

For n = 3



For n = 4



1. A brief discussion on why DFS Tree Search behaves differently from BFS in this specific puzzle.

The DFS Tree search behaves differently due to lack of completeness and redundancy management. The DFS tree search does not maintain a visited or closed set. Moving left and then again right brings us to the same position in the puzzle so DFS is prone to stuck in an infinite loop or keep on exploring the already visited nodes.

2. An explanation of what happened to the success rate when moving from  $n = 3$  to  $n = 4$ .

The number of possible states increase exponentially, for the 8-puzzle problem there are  $9! = 3,62,880$ , while for the 15-puzzle problem the number jumps highly to  $16! = 2 \times 10^{13}$  which is increased by several order.

The success rate of DFS, DLS, and Iterative deepening is already 0%, while BFS has some success rate and finds solution in approx. 0.5 second.

But as we move to  $n = 4$ , most likely the success rate fall to 0% for the 15-puzzle problem for all the 4 algorithms.